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## USN-2VX24UE023

## Exp.No- 7(B)

## Title- 0/1 Knapsack Problem using Greedy Method

In [ ]:

```
# Taking inputs from user
n = int(input("Enter the number of items: "))

print("Enter the weight and profit for each item:")
weight = []
profit = []
for i in range(n):
    w = int(input("Weight of item {}: ".format(i+1)))
    p = int(input("Profit of item {}: ".format(i+1)))
    weight.append(w)
    profit.append(p)

capacity = int(input("Enter the capacity of the Knapsack: "))

def greedy_knapsack(capacity, weight, profit, n):
    # Compute the profit-to-weight ratio for each item
    ratio = [(profit[i] / weight[i], weight[i], profit[i]) for i in range(n)]

    # Sort items by profit-to-weight ratio in descending order
    ratio.sort(reverse=True, key=lambda x: x[0])

    total_profit = 0
    current_weight = 0

    for ratio_value, w, p in ratio:
        if current_weight + w <= capacity:
            total_profit = total_profit + p
            current_weight = current_weight + w
        else:
            remaining_capacity = capacity - current_weight
            total_profit = total_profit + ratio_value * remaining_capacity
            break

    return total_profit

max_profit = greedy_knapsack(capacity, weight, profit, n)
print("====================================================")
print("Maximum Profit Earned (Greedy Approach) = ", max_profit)
print("====================================================")
```

In [ ]:

```
Enter the number of items:  3
Enter the weight and profit for each item:
Weight of item 1:  10
Profit of item 1:  60
Weight of item 2:  20
Profit of item 2:  100
Weight of item 3:  30
Profit of item 3:  120
Enter the capacity of the Knapsack:  40
====================================================
Maximum Profit Earned (Greedy Approach) =  200.0
====================================================
```